



Growing AI Like A Child

<https://growing-ai-like-a-child.github.io/>

Vision Language Models See What You Want but not What You See

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Introduction

Knowing others' intentions and taking others' perspectives are two core components of human intelligence that are considered to be instantiations of theory-of-mind. Infiltrating machines with these abilities is an important step towards building human-level artificial intelligence. Here, to investigate **intentionality understanding** and **level-2 perspective-taking** in Vision Language Models (VLMs), we constructed the IntentBench and PerspectBench, which together contains over 300 cognitive experiments grounded in real-world scenarios and classic cognitive tasks. We found **VLMs achieving high performance on intentionality understanding but low performance on level-2 perspective-taking**. This suggests a potential dissociation between simulation and theory-based theory-of-mind abilities in VLMs, highlighting the concern that they are not capable of using model-based reasoning to infer others' mental states.

Intentionality Understanding

Intentionality understanding is the capacity to comprehend the **mental content for action** in another mind. In developmental psychology, a critical subset of intentionality understanding experiments involves tests of action understanding. Typically, cartoon stimuli built via physic simulation engine are used frequently in action understanding in developmental psychology. However, a common critique of cognitive psychology tasks is that they lack realism and have limited applicability to real-world situations. Here we incorporated **real-world ambiguous scenarios** into IntentBench for explicitly testing intentionality understanding in ethological conditions.

Perspective-taking

Level-2 perspective-taking involves the understanding of how another person may **see the same thing differently**. Despite its relatively low level in the perspective-taking hierarchy, this ability requires model-based reasoning, exemplified in the visual domain as inferences based on **mental rotation**. We develop the **Three Mountain task** into formats that are suitable for benchmarks with minimal confounding details while preserving real-life spatiality. In particular, we use groups of 3-4 commonly-seen elastic cans organized into different spatial patterns to mimic the mountain model. Like in the original task, we use a doll placed to face the organization from different angles as the object of perspective-taking.

A



What do the people in the picture intend to do?

A. To test quality of the fence
B. To test quality of the ladder
C. To climb higher to repair the room

Correct Answer: C

GPT4o: C The image shows a person standing on a ladder that has been propped up against the railing of a higher level of a building it looks like the person on the ladder is attempting to reach and work on the roof area, which suggests they are likely conducting a repair or maintenance task this situation does not reflect the testing of the fence or the ladder",

B



What do the people in the picture intend to do?

A. To sit on sofa
B. To change a lightbulb
C. To climb on a sofa
D. To climb down from ceiling to a sofa

Correct Answer: B

GPT4o: B The person in the picture is standing on a sofa that has been precariously positioned vertically they are reaching up and holding a light fixture, suggesting they are in the process of changing the lightbulb

C



What do the people in the picture intend to do?

A. To prevent the mini fridge from slipping off while driving
B. To stay on the roof
C. To kick the mini fridge
D. To move the fridge down from the roof

Correct Answer: A

GPT4o: A The person is seen holding onto a mini fridge on the roof of the car this indicates that their purpose is to stabilize and secure the fridge, preventing it from falling off while the vehicle is in motion

D



What does the man at the bottom intend to do?

A. To take the shirt off
B. To walk on the rooftop
C. To climb to the top floor
D. To do some construction work

Correct Answer: D

GPT4o: D The man at the bottom appears to be involved in some construction-related work due to his presence on a construction site and being at the edge of a high structure the way he is positioned, with other workers assisting him, suggests that he is trying to perform a task that requires him to be at that particular edge however, the manner in which he is being held and the precarious nature of his position indicate that he is performing this task in a very dangerous and unsafe way

E



What does the person in the picture intend to do?

A. To check the lighting bulb
B. To stand high with feet against the wall
C. To get off from the ceiling
D. To paint the ceiling

Correct Answer: A

GPT4o: A The person is balancing with their legs against the walls while reaching up towards the ceiling light fixture this position suggests they are in the process of inspecting, replacing, or fixing the lighting bulb

F



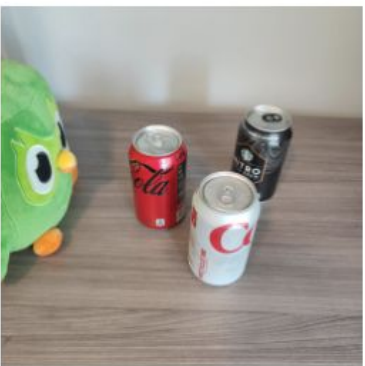
What does the person in the picture intend to do?

A. To keep the blood off his face
B. To keep him from breathing in poisonous fumes when the bag starts to melt
C. Both A and B
D. To protect him from the flying spark

Correct Answer: D

GPT4o: D The man appears to have placed a plastic bag over his head while using a grinder, seemingly to shield his face from the sparks produced by the tool however, this is extremely unsafe and not recommended because the plastic bag does not provide adequate protection against sparks, and it poses a serious risk of suffocation proper safety gear, such as a face shield, goggles, and a respirator designed for such tasks would be much safer and more effective in protecting him

A



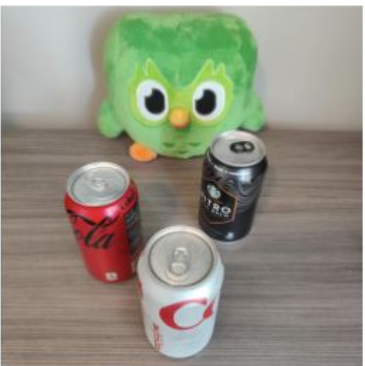
The image depicts a doll sitting on the table along with three cans in different colors: red, black, and silver. Please answer: from the doll's point of view, which can appears to be the rightmost of the three? Please select from the following choices and provide an explanation:

A. The red can
B. The silver can
C. The black can

Correct Answer: B

GPT4o: C "C the black can. From the doll's point of view, the black can appears to be the rightmost of the three cans on the table"

B



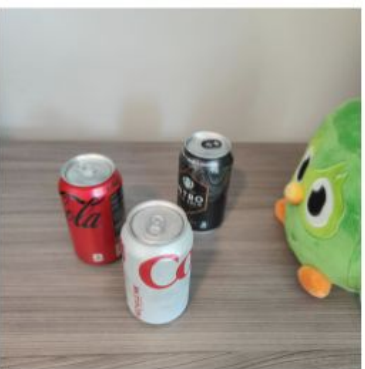
The image depicts a doll sitting on the table along with three cans in different colors: red, black, and silver. Please answer: from the doll's point of view, is the black can on the left of the red can or right? Please select from the following choices and provide an explanation:

A. Left
B. Right

Correct Answer: A

GPT4o: B "the answer is B. From the doll's point of view, the black can is to the right of the red can even when the perspective is flipped, the positions remain consistent relative to the doll's viewpoint"

C



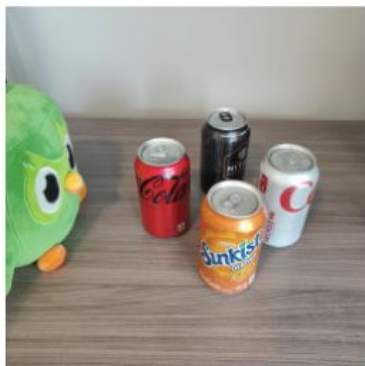
The image depicts a doll sitting on the table along with three cans in different colors: red, black, and silver. Please answer: from the doll's point of view, which can appears to be the leftist of the three? Please select from the following choices and provide an explanation:

A. The red can
B. The silver can
C. The black can

Correct Answer: B

GPT4o: A "The correct answer is A the red can. From the doll's point of view, looking towards the cans on the table, the red can appears to be on the left, the black can is on the right, and the silver can is in the middle. Therefore, the red can is the leftmost can"

D



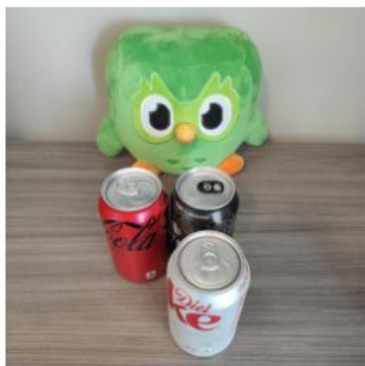
The image depicts a doll sitting on the table along with four cans in different colors: red, orange, silver, and black. Please answer: from the doll's point of view, which can appears to be the rightmost among all the cans it can see? Please select from the following choices and provide an explanation:

A. The red can
B. The orange can
C. The silver can
D. The black can

Correct Answer: B

GPT4o: C "The correct choice is C. From the doll's point of view, the silver can (on the right in the image) appears to be the rightmost among all the cans it can see\n\nso the answer is: c the silver can"

E

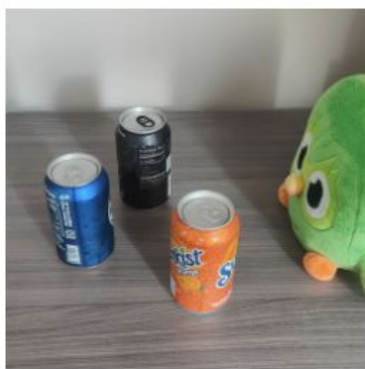


The image depicts a doll sitting on the table along with three cans in different colors: red, black, and silver. Please answer: how many cans can the doll see? Please answer with a number and provide an explanation.

Correct Answer: 2

GPT4o: 3 "The doll can see 3 cans. The cans are positioned in front of the doll, and since the doll's eyes are facing forward, it can see all three cans placed in view"

F



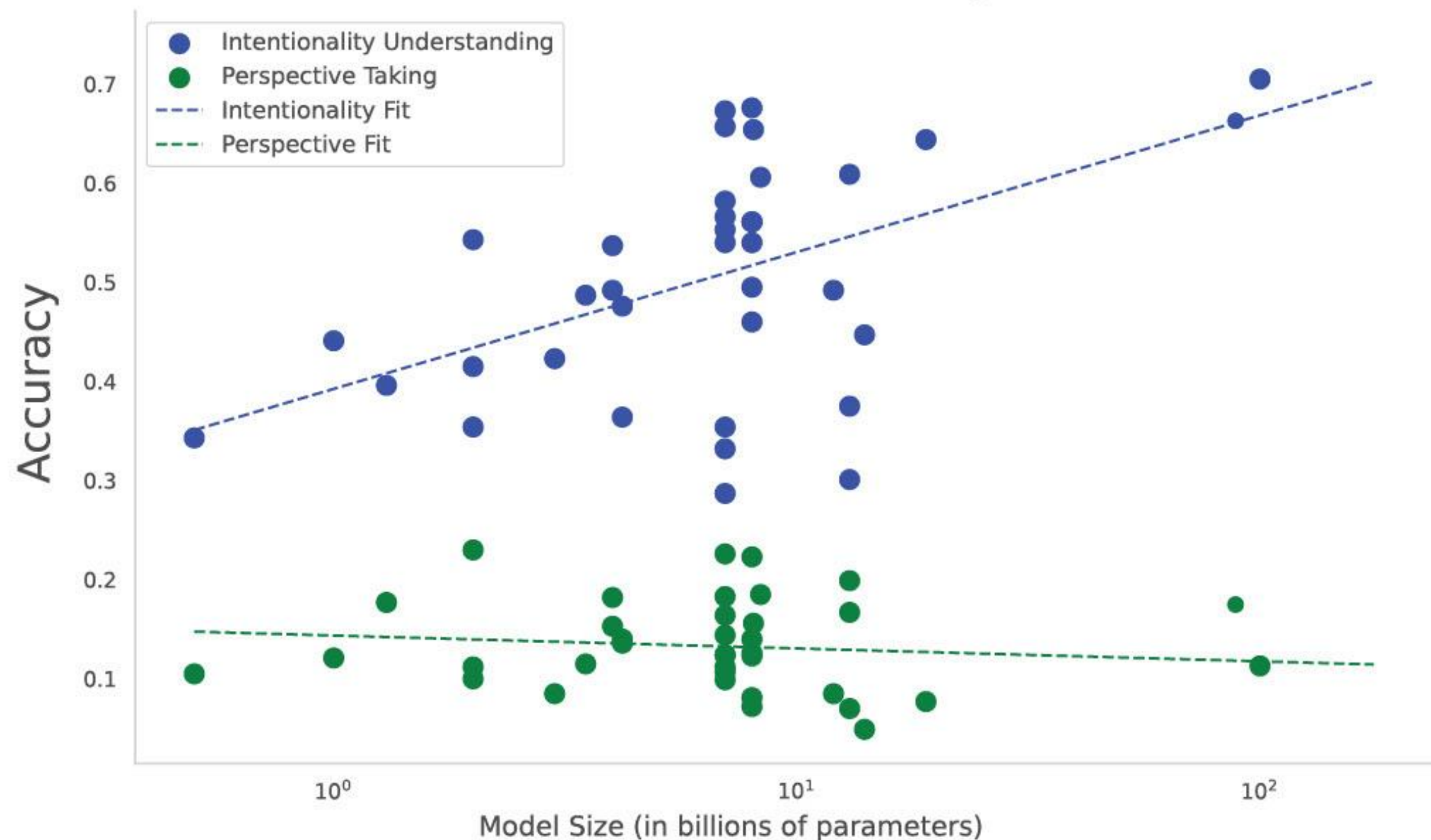
The image depicts a doll sitting on the table along with three cans in different colors: red, black, and silver. Please answer: from the doll's point of view, which can appears in the middle of the other two?" Please select from the following choices and provide an explanation:

A. The blue can
B. The black can
C. The orange can

Correct Answer: A

GPT4o: C "C the orange can. Explanation: from the doll's point of view, the orange can is positioned between the blue can (to the doll's left) and the black can (to the doll's right), making it appear in the middle of the other two cans"

Model Size vs Accuracy



Intentionality Understanding Improves with scale, Perspective-taking does not

We find a **clear dissociation** between the model performance on intentionality understanding and level-2 perspective-taking. Specifically, **all assessed models perform much better on IntentBench as compared to PerspectBench**. While some of the highest-performing models on IntentBench (e.g. GPT-4o) are able to reach near-human performance in intentionality understanding, they still perform poorly at level-2 perspective-taking. Even more strikingly, model performance on IntentBench exhibits a positive linear correlation with model size, whereas **model performance on PerspectBench do not improve alongside model size**.

This resonates with the concern that VLMs do not possess **internal models for reasoning** or at least cannot use them to take others' perspectives.